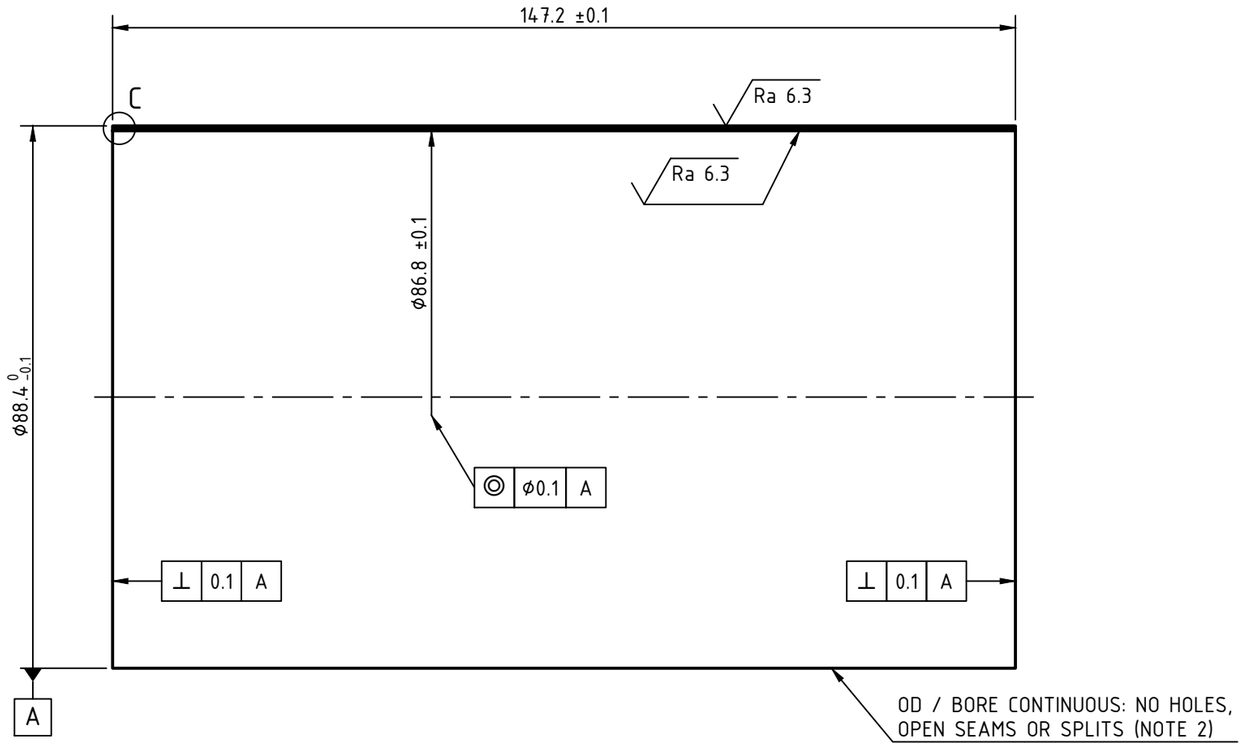

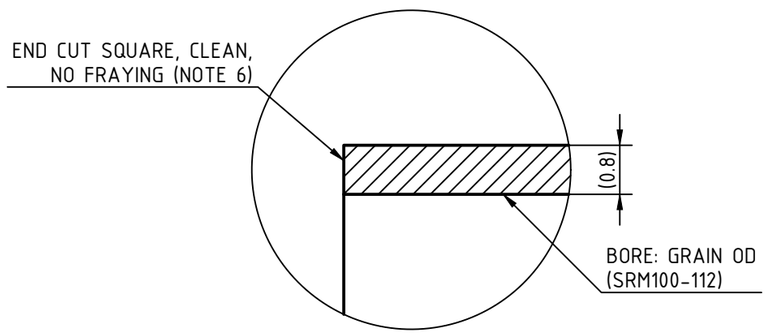

C (10:1)

NOTES

1. Material: spiral-wound kraft paper tube, plain kraft (not phenolic-impregnated), wall 0.8, density approx. 0.95 g/cm³. Stock: tube bought to the drawn bore and wall, cut to length. Plies bonded over their full area. Certificate of conformity required.
2. Propellant grain SRM100-112 is cast into this tube and stays in it: the tube is the grain's OD inhibitor. It must be continuous: no holes, open or overlapping seams, splits, blisters, delamination, dents or crushed ends. Inspect against a light.
3. Bore clean and dry: free of oil, wax, dust and loose fibres. OD free of raised seams and adhesive lumps.
4. Datum A is the OD axis. Sizes are two-point sizes at 20 °C on a tube conditioned dry (sealed with desiccant ≥ 24 h): kraft swells with humidity. Store sealed.
5. Length = grain length. In the finished segment the tube ends are flush with the grain end faces within 0.1 (SRM100-112).
6. Edges drawn sharp: ends cut square and clean, edge break 0.2 max, no burr or fraying. The ends bear on the grain spacers.
7. OD, also after the grain is cast in it, must pass freely over its full length through a ø88.60 ring gauge (liner SRM100-110 bore min).
8. Inspect 100 %: OD and bore at both ends, 2 directions 90° apart; length at 4 positions 90° apart; visual inside and out. Record per batch.
9. Mark SRM100-111 and batch on the OD within 20 of one end: indelible ink, 3 mm characters. The grain marking (SRM100-112) goes on the same OD.
10. Used on SRM100-000 item 9 (4 off), one per grain SRM100-112 (item 10). Single use: replaced with its grain after every firing (README, After every firing).

FIT AND POSITION

ITEM	VALUE
Liner SRM100-110 bore	ø88.60 … 88.70
Tube OD in liner bore	clearance 0.20 … 0.40 (diametral)
Case bore SRM100-101	no contact: the liner wall is between
Bore = grain SRM100-112 OD	ø86.8 ±0.1, cast in place
Aft face, motor z (tubes 1 … 4)	46 / 196.2 / 346.4 / 496.6
Ends bear on	grain spacers SRM100-109

Ra 6.3 (√)

MATERIAL: Spiral-wound kraft paper tube stock ID 86.8 x 0.8 wall	MASS: 30.8 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: Ra 6.3	EDGES: ISO 13715 -0.2	
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm	
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: CASTING TUBE — Grain casting tube / OD inhibitor, ø88.4 / ø86.8 x 147.2 - kraft				
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-111	REV: A	SHEET: 1 / 1